

SmartSense Portal UX Review

Twenty-six observations from the walkthrough, grouped and turned into work. Most are ours to fix in the rebuild. Six need a Product Owner decision, and three of those touch something already decided.

REVIEWED	NUMBERING	OURS TO FIX
4 September 2026	Follows the walkthrough, notes 2 to 27	17
NEEDS A DECISION		
6		

1 Design system	5 Answered from source
2 Density and hierarchy	6 Needs a decision
3 Navigation and interaction	7 What we take forward
4 Showing the data	

1 Design system

Four notes, all ours, all cheap to fix once because they are token and component problems rather than page problems.

NOTE 3 · THE CONTRAST FINDING IS MORE SPECIFIC THAN IT LOOKS

We measured the palette. The four text tiers all clear WCAG AA on every surface: 16.6, 10.7, 6.5 and 4.8 to one. On paper the type is fine, and the source comments say as much.

The structure is not. Every border token composites to between **1.1 and 1.9 to one** against its ground, in both themes, where the threshold for non-text UI is 3.0. Table row separators, card edges, input outlines and the panel boundaries that carry the whole layout are effectively invisible. That is what makes it feel bad: the words are legible and the shapes holding them are not.

Two supporting problems. The two faintest text tiers sit only 1.35 to one apart from each other, so a four-tier hierarchy reads as two. And `--brand` at #e63946 is the one colour that fails AA outright, at 4.33 on the surface it is most often painted on.

3	Colour contrast	OURS
	Rebuild the border scale against the 3.0 threshold rather than by eye, spread the four text tiers over a wider range or cut them to three, and either darken the brand red or stop using it on small text. Everything else in the palette holds up.	
7	Filter controls differ between Fleet and RV Owners	OURS
	One shows checkboxes, the other does not, for the same kind of multi-select facet. Both surfaces run on the same query specification with the same facet shape, so this is one component rendered two ways. One filter component, used everywhere.	
8	Gateways mixes percentages and ratios	OURS
	Either reads fine on its own. Pick one and apply it, with the other available on hover where the underlying count matters. Same fix as note 10.	
26	Small layout problems in places	OURS
	Noted, not itemised. These disappear in the rebuild rather than being worth cataloguing against a prototype we are replacing.	

2 Density and hierarchy

The strongest theme in the walkthrough. Five notes that are really one problem: the Portal explains itself in prose where it should show state.

2	Too much text on the landing page	OURS
	Agreed, and the package already went partway: the R2 branch we asked for in the handoff review did a two-tier disclosure pass, one question per page title with the provenance prose moved behind a top-bar control. That is close to what you are describing. We should see that branch before redesigning the same thing twice. Our position for the rebuild: the page states what is true, and the reasoning lives behind an affordance. Tooltip for a sentence, a panel for a paragraph, a separate page only when it is genuinely reference material.	
6	Overview is crowded and repeats itself	OURS
	Part of the repetition is structural: the Needs Attention section lists the same population as Fleet filtered to Needs Attention, which the Product Owner review already flagged as an open question about what the section is for. Worth settling that before laying the page out again.	

19	Relationship summary is cluttered	OURS
	Same fix as note 2. The relationships are a small set of facts and they should read as a set, not as a paragraph.	
10	Numbers should be consistent, and tooltips would help	OURS
	Two things underneath. Format consistency is ours. But some of the apparent inconsistency is real: authorized population, matched count and returned count are three different numbers the query contract deliberately keeps apart, and the page currently shows them without saying which is which. A tooltip that names the number is the right answer and it is cheap.	
9	Two search bars on a page	OURS
	By design, and the design is not obvious. The shell search is global across VIN, RV model and Gateway, server-scoped and field-authorized. The page search filters the population you are looking at. Two different things wearing the same control. Our fix is to make them look different rather than to remove one: the global search belongs in the top bar as a command affordance, the page filter belongs in the table toolbar and should say what it filters.	
3	Navigation and interaction	
4	Overview's quick filters could appear on Fleet too	OURS
	Fleet already carries the same six facets, so this is really about the one-click path: Overview offers a pre-set entry that Fleet makes you assemble. The Product Owner disposition for the Needs Attention section already asks for Open in Fleet to carry its filter context, which is the same idea. Build that path once and use it from both.	
18	Needs Attention navigation should focus the actual problem	OURS
	Agreed, and it is more than navigation. Needs Attention is a union over four dimensions: any active condition, reporting overdue, reporting unknown, or configuration drift or failed. Landing on a VIN without saying which of the four put it there is the gap. Carry the reason through the click and open the subpage that answers it.	
13	Fields in the detail side panel should be interactive	OURS
	Clicking a location to see it on a map, a gateway id to open the gateway, a definition to open the definition. This is the same request as note 14 and we would build one pattern for both: a field that names another object is a link to it, and a field with a spatial value gets a viewer.	

16	Vehicle history needs a date range, not just presets	SCOPED
	Already agreed and already scoped. The Product Owner tranche defers a Last 24h / 7d / 30d / 90d / Custom selector filtering canonical event time, and the Significant-G range control on Overview was built that way. This is finishing the job rather than a new request.	
21	Pages have no calls to action, only data	OURS
	True, and partly deliberate. The Alerts destination is evidence and read-only by decision, no Portal resolve or clear. Device commands and Gateway Settings are held at the service, so no control for them exists anywhere. Those two should stay as they are. What is genuinely missing is everything else: no export where export is authorized, no way to act on a Needs Attention row, no path from an alert event to the definition that produced it. We would go through the register surface by surface and name the action each one should offer, then check it against the page policy.	
4	Showing the data	
11	Sensor charts should compare series	OURS
	Humidity and temperature side by side, or overlaid on one plot with two axes. The chart contract already returns prepared series with per-series aggregates, so multi-series is a presentation change rather than a backend one. Worth doing.	
14	Vehicle history should show a map	OURS
	Coordinates are already on the rows and the geofence editor already carries a map component. Marking the event location when something goes wrong is the right idea. One constraint: precise coordinates are gated by <code>location.precise.view</code> , so the map has to degrade to a withheld state rather than fall back to an approximate pin.	
20	Gateway detail gives no freshness or judgement	OURS
	Three separate gaps and all three are real. Whether the page polls or streams is our architecture decision and we should state it on the page. Last updated is missing and should never be missing on a live surface. And a battery reading with no context is a number, not a status. The third one needs care: turning a value into good or bad is a judgement, and severity in this product is server-authoritative and never derived by the client. So the threshold has to come from the server, not from the page.	

23 **GPS loss should show the last known position, and alert types could have their own presentation**

OURS

Both good. A connection-loss alert whose whole point is that position is unknown should show where it was last known, with the age of that fix. And a tyre pressure alert, a Significant-G event and a gateway offline alert have almost nothing in common beyond a timestamp, so one generic layout serves none of them well.

We would define a small set of event presentations keyed to the measurement rather than one template with optional fields.

24 **Geofence: drawing an area on the map could be better**

OURS

The editor already does click to place, drag the circle centre, drag and insert and delete polygon vertices, translate and scale. So the capability is there and the flow is what needs work. Worth a proper design pass rather than a fix.

The separate geometry question is in section 6, because geometry is currently a filter dimension on the sites list rather than a page, and we want to check what you were looking at.

5 **Answered from source**

12 **What is the difference between Readings and Raw?**

ANSWERED

Readings are governed field values: the measurement after conversion, in its display unit, from the datapoint catalogue. Raw is the pre-conversion device value, including raw accelerometer axes, and it is the only route in the whole registry that is Read for SmartSense Administrator and No Access for the other four profiles.

So the split is a privilege boundary, not a data boundary. The distinction is invisible on the page, which is why the question came up. Naming it is a copy fix.

6 **Needs a decision**

Six. Three of them collide with something already decided, so they cannot simply be built.

5	Trend graphs for Reporting status and Active conditions	DECISION
This is the one to be careful with. The Overview once had a 24h / 7d / 30d control that applied no window to any figure, and when it was removed the disposition was explicit that no replacement historical-health behaviour was invented, and QA asserts no health-trend concept appeared. So a trend graph is not a presentation change. It is the historical-health capability that was deliberately left out, and it needs a definition first: trended against what, event time or report time or condition onset, and over which population.		
15	Green, amber and red coding on history values	DECISION
Same shape of problem as note 20. Colouring a value implies a threshold, and thresholds here are governed configuration, not presentation. Once the definitions exist we can colour against them. Until then, colouring would be the Portal inventing a judgement.		
17	Should capabilities be hidden on vehicle history?	DECISION
Genuine question and we lean the same way you do. Capabilities explain what a unit can report, which is exactly the context needed when a field is empty. Behind a toggle it is discoverable only if you already knew to look.		
24	Do we need a separate geometry page?	CHECK
In the current source, geometry is a filter dimension on the sites list, not a page of its own. We want to check which surface you meant before proposing a change.		
22	An RV illustration with data layered onto it	SCOPE IT
The strongest idea in the list and the one most worth exploring. Six tyre positions with canonical names and aliases already exist per model, so an alert already knows which wheel it belongs to. Putting that on a vehicle diagram rather than in a table row is a real improvement, and it is the same picture J3309 asks the mobile HMI to show. It is a design investment rather than a fix, and it should be scoped deliberately. Our suggestion is to prototype it against the VIN detail surface first, where the position data is richest.		

27 How dynamic are subscription plans, exactly?

DECISION

Nothing in the package answers this. Subscriptions and Subscription Plans are deferred with no feature work approved, and the only fixed points we have are that 36 months is the controlled default included term and that the permitted range is an open decision. Anything beyond that is undefined.

Worth thinking about before it is asked for, not before it is scoped.

7 What we take forward

Three of these change how we sequence the work rather than just what we build.

ITEM	EFFECT
Border and hierarchy scale note 3	Goes into the design tokens at the start of S1, not into a polish pass. Every component inherits it, so fixing it later means touching everything.
One filter component notes 4, 7, 9	Built once alongside the windowed grid in S1. Eleven surfaces use it, and the inconsistency in the prototype is what happens when it is not one component.
Event presentation per type notes 22, 23	Affects the alert event contract we are proposing, since a type-specific presentation needs the event to carry enough to drive it. Worth settling before S6.

The rest are ordinary work inside the stages they belong to. Note 25's observation that the alert definition flow resembles the Foodie wizard is worth acting on internally: that pattern is already built and reviewed, and reusing its structure would save the S6 authoring surface from being designed from nothing.

Internal working document. Observations from the walkthrough of the Portal prototype, numbered as recorded. Contrast figures computed from the theme tokens in the prototype source against WCAG 2.1 AA. Where a note collides with a recorded Product Owner decision, that decision is named rather than overridden.